

AMMAAR SAADAT

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Education & Awards

University of Michigan - Ann Arbor

Bachelor of Engineering - Computer Science and Engineering
Dean's List - Honors Recognition

May 2025

Ann Arbor, Michigan

Michigander Scholars in Semiconductors Scholarship

Awarded \$5,000 for academic excellence to Michigan's semiconductor industry

Jan. 2025

Ann Arbor, Michigan

Courses: Operating Systems, Linear Algebra, Computer Security, User Interface Development, Web Systems, Computer Architecture, Info Retrieval, Human-Centered Software, Data Driven Software, Calc I-III

Experience

Sarf

Aug. - Dec. 2025

Software Engineer — Next.js, React, iOS, Android, SupabaseSQL

New York City, New York

- Designed a scalable platform using SQL database models with row-level security policies, client/server state synchronization for dynamic rendering, and auth middleware for verification, **resulting in up to \$200,000 funds raised, 50+ early sign-ups, 5 investor pitches, and moved product toward patent-pending status.**
- Developed core product infrastructure such as user accounts, live market data ingestion pipelines, AI itinerary processing chain, payment flow, and a ledger system

KLA

May - Aug. 2024

Software Engineering Intern — gRPC, Protobuf, Java, SpringBoot

Ann Arbor, Michigan

- Developed a scalable layered architecture by integrating a gRPC microservice for remote internal tool access, **reducing internal tool interaction time by 85% across 2+ engineering divisions**
- Designed 8+ low-latency RPC endpoints with Protobuf serialization and client/server stub dependency generation for decoupling integrations — standardizing plug-in interfaces and enabling global deployment in international labs

Projects and Entrepreneurship

Multi-Threaded Network File Server | *Networking, File Systems, C++, Threads*

Apr. 2025

- Designed a thread-per-request file server by introducing per-file locking and hand-over-hand traversal for safe parallel access to shared file system state across concurrent TCP client connections, **leading to 95% data validation compliance**
- Ensured fault tolerance of **1,000+ concurrent client commands** by implementing sanitization of malformed input, preventing race conditions with synchronization, and enabling disk crash consistency, verified through stress tests and I/O failure edge-case analysis

Multicycle CPU Simulator | *C, Assembly, LC2k, ARM*

Apr. 2024

- Developed a multicycle CPU and unified L2 cache simulator in C for the LC2K architecture, implementing configurable block size, set associativity, and LRU replacement; supported write-back and write-allocate policies, achieving optimizations across 20+ assembly test programs with detailed memory transfer tracking, and hazard optimizations

Takhat Cafe | *Co-Founder | Project Management, Leadership*

Mar. 2025 - Present

- Led marketing team** by advertising on social media, community events, and designing flyers to reach 5k unique accounts in 3 months, **resulting in 1000+ sales.**
- Secured 10+ clients by pitching services** and adjusting based on clientele feedback, resulting in **1.2k units sold in 6+ cities**

Vitra.ai | *Co-Founder | Gemini, Python, Flask, React, Project Management*

May 2025

- Co-founded Vitra, a non-profit **AI transliteration app** to reduce diagnosis time and increase accuracy of patients different language barriers via **40+ languages** for real-time communication
- Scaled ideation to an MVP with **demos of 2 clinics** by assembling and managing a team of undergraduate engineers using agile workflows, Gantt charts, and phased prototyping to demo

Technical Skills and Interests

Languages: Python, C++, Bash, Java, C, JavaScript, Swift, Flutter, HTML/CSS5

Frameworks: TypeScript, Gemini, OpenAI, Pinecone, React, SQL, gRPC, Next.js, Flask

Interests: Raspberry Pi, Volleyball, Marketing, Hiking, Gardening, Pokemon Go